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# End of Year Project

## Second Year Engineering Cycle

Specialty: Computer Networks and Telecommunications

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# Intelligent Multimodal approach for Automatic Generation of High-Impact Short Formats (Shorts / Reels)

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Defended on june 2026 in front of a jury composed of:

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# Acknowledgments

# Abstract

Write your abstract here.

**Keywords: ..., ... .**

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# List of Acronyms

- **Abbrev.** Abbreviation

# List of Symbols

| | |
|---|---|
| $T$ | Number of frames per segment |
| $C$ | Number of colour channels |
| $H, W$ | Spatial dimensions of each frame (px) |
| $\mathbf{v}_i, \mathbf{a}_i, \mathbf{t}_i$ | Vision, audio, and text embeddings for segment $i$ |
| $N$ | Total number of video segments |

# Introduction

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# 1

# State of the Art

## Introduction

Video understanding is an inherently multimodal problem. A highlight is rarely defined by a single signal  it emerges from the alignment, or sudden divergence, of what is seen, heard, and spoken simultaneously. Building a system capable of detecting such moments therefore requires models that can reason jointly across modalities, not process them in isolation. This chapter traces the progression from unimodal approaches and their limitations, through fusion-based systems, to joint multimodal representation learning  the paradigm this work builds upon. Each step in this progression is motivated by the shortcomings of the one preceding it, converging on the architectural foundation of the proposed approach.

# 1 Multimodal Video Analysis

## 1.1 The Complementary Nature of Vision, Audio, and Text

## 1.2 Modality Contributions in Video Understanding Tasks

## 1.3 Domain Dependency of Modality Relevance

## 1.4 Highlight Detection as a Multimodal Task

# 2 Unimodal Approaches for Video Understanding

Video understanding has traditionally been approached through unimodal analysis, where only a single modalityvisual, audio, or textis processed independently. While such approaches can extract meaningful information, they often fail to capture the full semantic and emotional richness of long-form videos.

For highlight detection and segmentation tasks, relying on a single modality may lead to incomplete or misleading interpretations of what constitutes a high-impact segment.

## 2.1 Visual Understanding

Visual-based methods focus on analyzing video frames to extract spatial and temporal patterns. Early systems relied on handcrafted features, while modern approaches use deep convolutional neural networks to learn visual representations automatically.

Typical visual analysis techniques include:

- Frame-level feature extraction

- Optical flow for motion estimation

- Shot boundary detection

- Scene classification

- Object and action recognition

Libraries such as OpenCV and Decord are widely used for frame extraction and temporal sampling in video processing pipelines. However, visual saliency does not necessarily correspond to semantic importance. A visually dynamic segment may lack narrative relevance, while a static scene containing crucial speech content may represent a key moment. Therefore, visual-only segmentation is insufficient for detecting meaningful high-impact clips.

## 2.2 Audio Understanding

Audio-based approaches analyze the acoustic signal to detect speech, music, sound events, and emotional cues. These systems typically rely on spectral and temporal representations of sound.

Common techniques include:

- Mel-Frequency Cepstral Coefficients (MFCCs)

- Spectrogram analysis

- Energy-based segmentation

- Pitch and prosodic feature extraction

- Speaker diarization

The Python library Librosa is commonly used for extracting spectral and rhythmic features such as MFCCs, chroma features, spectral centroid, and tempo descriptors.

Although audio intensity can signal emotionally charged moments, it may also produce false positives due to background music or noise. Furthermore, impactful segments may not always correspond to peaks in acoustic energy. Hence, audio-only systems lack contextual grounding.

## 2.3 Textual Understanding

Text-based video understanding relies on Automatic Speech Recognition (ASR) to convert spoken content into transcripts, enabling Natural Language Processing (NLP) analysis.

Modern ASR models such as Whisper provide robust transcription for long-form and multilingual audio content.

Once transcribed, NLP techniques can be applied for:

- Keyword extraction

- Topic segmentation

- Sentiment analysis

- Emotion detection

- Semantic similarity analysis

Textual analysis enables identification of semantically important moments in videos. However, it ignores visual expressiveness, cinematic effects, and acoustic intensity, which are often crucial in determining whether a segment is engaging or impactful.

## 2.4 Representative Unimodal Models and Frameworks

### 2.4.1 Whisper

Whisper is an automatic speech recognition (ASR) system developed by OpenAI. It is trained on 680,000 hours of multilingual and multitask supervised data collected from the web, enabling robust transcription and translation capabilities across multiple languages.

**Key Features:**

- Multilingual speech recognition supporting 99 languages
- Robust performance with various accents, background noise, and technical jargon
- Capable of translation from non-English languages to English
- Provides timestamped transcriptions suitable for subtitle generation

For short-form content generation, Whisper can automatically generate captions, subtitles, and transcripts from audio tracks, enabling better content accessibility and searchability.

### 2.4.2 Librosa

Librosa is a Python library for music and audio analysis. It provides the building blocks necessary to extract audio features, analyze sound content, and process audio signals for

machine learning applications.

**Key Features:**

- Audio feature extraction (MFCCs, chroma features, spectral contrast, zero-crossing rate)
- Beat and tempo tracking for rhythm analysis
- Onset detection for identifying note beginnings
- Audio visualization tools (spectrograms, waveforms)
- Support for various audio formats

In our project, Librosa enables audio preprocessing, feature extraction for content analysis, and rhythm-based editing for synchronizing visual cuts with music beats.

### 2.4.3 OpenCV and Decord

OpenCV (Open Source Computer Vision Library) and Decord are two essential frameworks for video processing and computer vision tasks.

#### 2.4.3.1 OpenCV

OpenCV is a comprehensive open-source computer vision and machine learning library with over 2,500 optimized algorithms.

**Key Features:**

- Video capture, processing, and writing
- Image and video filtering, transformation, and enhancement
- Feature detection and matching (SIFT, ORB, etc.)
- Object detection and tracking
- Real-time video analysis capabilities

#### 2.4.3.2 Decord

Decord is an efficient video reader library designed for deep learning workflows. It provides fast, GPU-accelerated video decoding and random frame access.

**Key Features:**

- High-performance video decoding
- Random frame sampling without sequential decoding
- GPU-accelerated processing
- Seamless integration with PyTorch and other deep learning frameworks

For short-form content generation, OpenCV provides essential video processing capabilities for editing, filtering, and visual effect application, while Decord enables efficient video loading and frame extraction for model training and inference.

### 2.4.4 Synthesis for Short-Form Content Generation

The unimodal frameworks presented above serve complementary roles in our automatic short-form content generation pipeline:

- **Whisper** provides transcription and captioning capabilities for audio content
- **Librosa** enables music analysis and beat-synchronized editing
- **OpenCV** delivers comprehensive video processing and visual effect capabilities
- **Decord** ensures efficient video loading and frame extraction for model training

# 3 Limitations of Unimodal Approaches for Highlight detection

For the task of segmenting long videos into short, high-impact reels, unimodal systems face significant limitations:

- Visual systems detect motion but not meaning.

- Audio systems detect intensity but not narrative relevance.

- Text systems detect semantics but not emotional amplification.

High-impact segments typically emerge from the interaction between modalitiesfor example:

- A powerful statement (text) delivered with emotional tone (audio)

- A dramatic visual scene reinforced by intense background music

- A key announcement accompanied by audience reactions

Therefore, unimodal analysis fails to model cross-modal dependencies that define impactful video moments.

# 4 Fusion-Based Multimodal Systems

# 5 Joint Multimodal Representation Learning

Joint multimodal representation learning addresses a fundamental limitation shared by both unimodal and fusion-based approaches: the absence of a unified semantic space in which modalities such as vision, audio, and text can be collectively treated and compared. Rather than processing each signal independently and combining outputs,the joint approach trains a single model to vimap all modalities into a shared high-dimensional embedding space, where geometric proximity reflects semantic similarity regardless of the originating modality.
Modern multimodal systems are inspired by the contrastive learning framework popularized by CLIP-like architectures, enabling cross-modal reasoning where heterogeneous modalities are treated as different expressions of the same underlying objective.

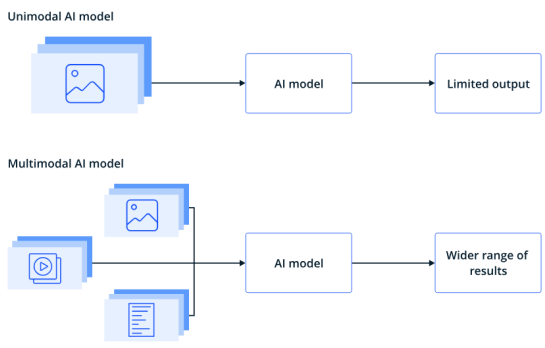


Figure 1.1: Comparison of unimodal and multimodal processing frameworks. Unimodal systems process a single data type, while multimodal systems integrate multiple modalities into a unified representation [AI Planet, 2024].

## 5.1 Architectural Overview

Although joint multimodal models vary in design, most share a common three-stage architecture: modality-specific encoders, a fusion mechanism, and optionally a decoder for generative tasks Sajid [2024].

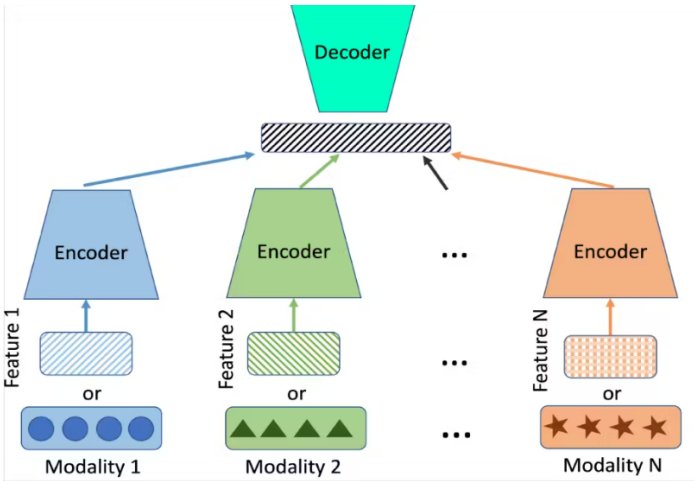


Figure 1.2: Joint multimodal model Architecture
Sajid [2024]

Modality-specific encoders transform raw inputs into dense feature vectors: image encoders employ **CNNs** or **Vision Transformers (ViT)**, text encoders leverage **transformer architectures** to map token sequences into contextual embeddings, and audio encoders such as Wav2Vec2 [Baevski et al., 2020] convert raw waveforms into representations capturing rhythm, tone, and semantic content. Once encoded, a fusion mechanism integrates these embeddings into a unified representation. Four strategies are commonly used: *early fusion* combines raw features before processing; *intermediate fusion* merges latent representations; *late fusion* combines final modality outputs; and *hybrid fusion* applies elements of all three at different stages. These strategies are implemented through attention-based methods Vaswani et al. [2017], feature concatenation, or dot-product interactions between modality vectors.

## 5.2 Contrastive Learning Framework

Contrastive learning is the fundamental training strategy underlying modern multimodal models. Rather than predicting explicit class labels, the model learns by comparing pairs of inputs: a **positive pair** consists of semantically aligned samples from different modalities — an image and its corresponding caption, or a video frame and its paired

audio segment — while a **negative pair** consists of mismatched samples. The model is trained to pull positive pairs closer together in the embedding space while pushing negative pairs apart, with the supervision signal arising entirely from cross-modal correspondence rather than manual annotation.

This is formalized through the InfoNCE objective [van den Oord et al., 2019]. Given a batch of $N$ matched pairs $(x_i^{(A)}, x_i^{(B)})$ from two modalities $A$ and $B$, each encoder maps its input to a **normalized** embedding such that the correct match is identifiable among all $N$ candidates. The symmetric loss is:

$$\mathcal{L} = -\frac{1}{2N} \sum_{i=1}^{N} \left[ \log \frac{e^{\mathrm{sim}(\mathbf{e}_i^A, \mathbf{e}_i^B)/\tau}}{\sum_j e^{\mathrm{sim}(\mathbf{e}_i^A, \mathbf{e}_j^B)/\tau}} + \log \frac{e^{\mathrm{sim}(\mathbf{e}_i^B, \mathbf{e}_i^A)/\tau}}{\sum_j e^{\mathrm{sim}(\mathbf{e}_i^B, \mathbf{e}_j^A)/\tau}} \right] \tag{1.1}$$

where $\mathrm{sim}(\cdot, \cdot)$ is cosine similarity and $\tau$ is a temperature hyperparameter controlling the sharpness of the similarity distribution. A lower $\tau$ sharpens the objective, producing a more discriminative but noise-sensitive embedding space; in practice, $\tau$ is either fixed — CLIP uses $\tau = 0.07$ [Radford et al., 2021] — or learned per modality, as in ImageBind [Girdhar et al., 2023]. This framework has proven highly scalable, enabling multimodal pretraining on large-scale web data without any manually labelled supervision.

## 5.3 Shared Embedding Spaces

The product of contrastive training is a shared embedding space, typically of dimension $d \in \{512, 768, 1024\}$, in which each modality is processed by a dedicated encoder — a transformer-based language encoder for text, a vision transformer for images, or a spectrogram-based encoder for audio [Sajid, 2024] — followed by a projection layer that maps the resulting features into the same latent dimensionality. All embeddings are constrained to lie on the unit hypersphere $\mathbb{S}^{d-1}$ via $\ell_2$ **normalization**, making cosine similarity a universal proximity measure across modalities:

$$\mathrm{sim}(\mathbf{e}^A, \mathbf{e}^B) = \mathbf{e}^A \cdot \mathbf{e}^B \qquad \text{since } \|\mathbf{e}\| = 1 \tag{1.2}$$

Within this shared space, semantically related inputs are positioned close to one another regardless of modality: an image of a cheering crowd, an audio clip of applause, and a textual description of excitement occupy nearby regions of the embedding manifold.

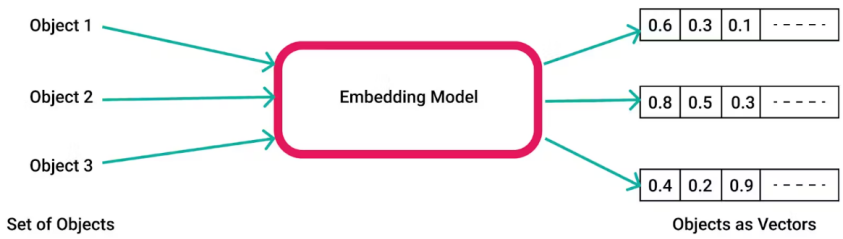


Figure 1.3: Multimodal embeddings: raw inputs from different modalitiesprojected into a common high-dimensional vector space Sajid [2024].

This unified representation facilitates cross-modal similarity computation and retrieval without requiring explicit modality-specific fusion mechanisms.

A further property is *emergent transitivity*: a model trained on image-audio and image-text pairs separately achieves implicit audio-text alignment through the shared visual anchor, without the two modalities ever being trained together directly.
ImageBind [Girdhar et al., 2023] exploits this property to bind six modalities — image, text, audio, depth, thermal, and IMU — using only image-paired data for each, achieving full cross-modal alignment across all modality combinations at inference time.

## 5.4 Zero-Shot Transfer and Cross-Modal Retrieval

The shared embedding space directly enables zero-shot transfer: the ability to perform recognition or scoring on novel concepts expressed as natural language, without any task-specific training. Because the model learns semantic alignment rather than task-specific boundaries, it can transfer to unseen categories without fine-tuning. A concept such as "crowd applause" is encoded by the text encoder into a semantic anchor embedding; candidate segments are then scored by their cosine similarity to this anchor, regardless of modality:

$$\text{score}(x) = \text{sim}(f_{\text{text}}(q),\ f_{\text{target}}(x)) \tag{1.3}$$

The shared space enables cross-modal retrieval: a natural language query can retrieve the most semantically relevant video segment purely on embedding proximity, while an audio clip of crowd cheering can retrieve the visual moment of a goal.
This "search by anything" capability supports text-to-image, image-to-text, audio-to-text, and audio-to-video retrieval within a single framework.

## 5.5 Advantages and Limitations of Joint Multimodal Learning

Joint multimodal learning paradigm offers four key advantages over alternative approaches.It provides **native cross-modal alignment**, to encode semantic correspondences directly in the geometry of the embedding space rather than through engineered fusion rules. It enables **zero-shot generalization**, allowing the system to handle new content categories at inference time through natural language, without depending on annotated datasets. It supports **modality-agnostic scoring**, where a single similarity function operates uniformly across all available modalities.

Finally, models (such as ImageBind) benefit from **large-scale pretraining** on internet-scale paired data, encoding world knowledge that transfers effectively to downstream tasks without fine-tuning.

Despite these strengths, the paradigm presents notable limitations. The most significant is the **absence of temporal modeling**: joint multimodal models produce a single embedding per fixed-length segment and cannot model sequential dynamics across segments. Narrative structures, such as the comedic buildup across several exchanges before a punchline, must be handled by external temporal analysis. A second limitation is **embedding norm uniformity**: because all embeddings are $\ell_2$-normalized, their magnitude carries no information, making raw energy-based scoring inapplicable without additional statistical processing. Finally, **modality imbalance** in pretraining data: image-text pairs, being far more abundant than audio-centric combinations, can reduce the discriminative power of audio embeddings in content where acoustic cues are the primary signal.

Addressing these challenges remains an active area of research, particularly for video-centric applications where temporal coherence and multimodal interaction play a central role.

# 6 Representative Multimodal Models

## 6.1 CLIP: Contrastive Language-Image Pre-training

CLIP (Contrastive Language-Image Pre-training) is a neural network developed by OpenAI that learns visual concepts from natural language supervision. It enables zero-shot transfer to arbitrary visual classification tasks without requiring task-specific training data.

### 6.1.1 Core Concept

CLIP is trained on a large dataset of text-image pairs collected from the internet. The model learns a joint embedding space where matching images and text descriptions are positioned close together. This is achieved through a contrastive objective that maximizes similarity between correct pairs while minimizing it for incorrect pairs.

### 6.1.2 Architecture

CLIP consists of two encoders:

- **Image Encoder:** A Vision Transformer (ViT) or ResNet that processes images.
- **Text Encoder:** A Transformer that processes text snippets.

Both encoders project their inputs into a shared embedding space, enabling direct comparison between visual and textual content.

### 6.1.3 Key Capabilities

- **Zero-Shot Classification:** CLIP can perform new visual classification tasks by simply providing the names of the categories in natural language, without any additional training.
- **Flexibility:** The model can handle diverse tasks including object recognition, action detection, geo-localization, and OCR.

- **Robustness:** CLIP reduces the "robustness gap" by up to 75% compared to traditional models trained directly on benchmarks.

### 6.1.4 Relevance to Our Project

For automatic short-form content generation, CLIP enables:

- Cross-modal search (finding videos or images using text descriptions)
- Automatic content tagging and categorization
- Quality assessment of generated content against textual descriptions
- Complementary capabilities to ImageBind, focusing on vision-language alignment

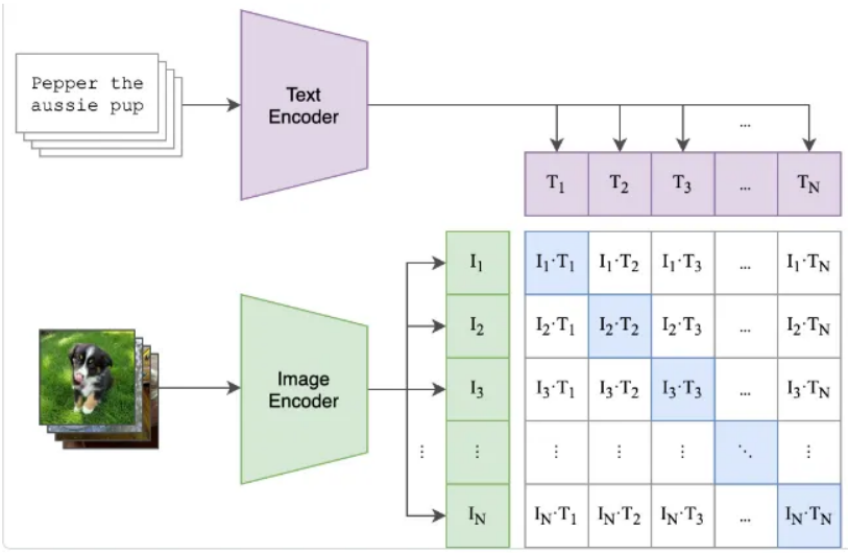


Figure 1.4: CLIP architecture with dual encoders for images and text

## 6.2 CLAP

Contrastive Language-Audio Pretraining (CLAP), as described in [Elizalde et al., 2022], is an extension of the contrastive learning paradigm of CLIP to the audio domain, with the aim of learning a joint representation space between audio and text to facilitate zero-shot audio classification and retrieval tasks. The architecture employs two encoders to process the audio and text modalities in the form of mel spectrogram and text, respectively, and is trained using symmetric contrastive loss to maximize the similarity between matching and minimize the similarity between non-matching audio and text pairs.

Despite its success in aligning the audio and text modalities, the limitations of the CLAP approach for the task of highlight detection in videos are as follows: the audio is processed in isolation without access to the visual information, thus making it impossible to leverage the audio-visual modalities. In addition, the encoders are optimized for short videos, thus limiting the modeling of long-range dependencies, which is critical in the case of videos. The use of architectures such as ImageBind, which combines the three modalities in a single representation space, can be more effective for multimodal video analysis.

## 6.3 Flamingo

The Flamingo model developed by DeepMind is a powerful family of Visual Language Models designed to process sequences of images or videos interleaved with text and generate free-form text as output.
Flamingo represents a notable progression in vision-language modeling by enabling a single model to perform multiple multimodal tasks with minimal task-specific fine-tuning.

A core concept of its architecture is the fusion of large, pre-trained, and frozen models specifically a **vision encoder** and a **language model**to leverage their existing knowledge without the need for intensive retraining .
To efficiently bridge these two modalities, Flamingo introduces **the Perceiver Resampler**, which compresses a huge number of visual features into a fixed-size collection of visual tokens , these tokens then condition the frozen language model through **Gated Cross-Attention layers**, which are interleaved between the original language model layers , these gated layers are essential for preserving the stability of the model during training and ensuring the language model's original performance remains intact at initialization.

A key contribution of Flamingo is its **few-shot learning** capability The model can adapt to various tasks with only a few input-output examples . Experimental evaluations showed strong performance across several visionlanguage benchmarks, such as multimodal reasoning tasks, visual question answering and image captioning [Alayrac et al., 2022].

## 6.4 LLAVA

LLaVA is a multimodal model designed to enhance general-purpose visual and language understanding through the use of visual instruction tuning [Liu et al., 2024].

The architecture of LLaVA consists of a **pretrained CLIP vision encoder** and a **Vicuna large language model**, which are connected by a two-layer **Multi-Layer Perceptron cross-modal projector** .
This MLP projector converts visual encoder tokens into post-embedding layer LLM tokens, replacing the simpler linear projection used in earlier versions to improve multimodal capabilities .
The training process follows a **two-stage pipeline:** an initial stage focused on modality alignment using roughly 600,000 image-text pairs, followed by a second stage of visual instruction fine-tuning where the model is trained on diverse conversation datasets .
This approach enables LLaVA to perform tasks such as image description, visual question answering, and multimodal dialogue generation. Compared to more complex architectures, LLaVA is considered more accessible and computationally efficient while maintaining competitive performance.

## 6.5 Video-LLaMA

Video-LLaMA [Zhang et al., 2023] is a video understanding model that uses the generative method of incorporating video content into large language models. Unlike embedding-based video retrieval models that treat text, vision, and audio as equivalent modalities in an embedding space, Video-LLaMA places the language model at the core of the model and treats vision and audio as conditionals for text generation. The model is instruction-tuned to process video content with synchronized audio for natural language reasoning on dynamic video content.

The model employs two branches: the Vision-Language branch that aggregates video features using Video Q-Former to extract temporal dependencies across frames using cross-frame attention mechanisms, and the Audio-Language branch that processes audio content using ImageBind [Girdhar et al., 2023] to extract features that can be used in conjunction with the language model.
Although Video-LLaMA is effective in processing temporal video content for scene un-

derstanding and question answering tasks, its autoregressive generation method is not appropriate for embedding-based video retrieval tasks. However, the training of video language models like Video-LLaMA is computationally expensive and demands large-scale video-text paired datasets, which limits its adoption by many research groups.

VideoLLaMA 2 [Cheng et al., 2024] is an improved version of VideoLLaMA that uses a spatial-temporal convolution connector instead of the dual branches of the original model to link the vision and audio models to the language model and improves audio content processing using BEATs [Chen et al., 2022].

## 6.6 ImageBind: A Foundational Model for Holistic Multimodal Integration

ImageBind, developed by Meta AI's FAIR Lab and open-sourced in May 2023, represents a paradigm shift in multimodal learning. It is the first AI model capable of binding information from six distinct modalities into a single, unified embedding space. For this project, which focuses on automatically generating high-impact short-form content (Shorts/Reels), ImageBind serves as a critical conceptual blueprint. It demonstrates how an AI system can move beyond processing disparate data streams to achieving a holistic understanding of content, mirroring the human ability to integrate sight, sound, and context simultaneously.

### 6.6.1 Core Concept: Joint Embedding Space

At its heart, ImageBind is built on the principle of creating a **joint embedding space**. This means that data from different modalities (e.g., a video clip, its background music, and a text caption) are mapped into a single, shared vector space. In this space, related concepts from different modalities are positioned close together. For a project like ours, this is a crucial capability, as it allows a system to directly compare, retrieve, and combine elements across audio, video, and text to generate a cohesive final product.

### 6.6.2 Modalities Integrated: A Sensory Spectrum for Content Creation

ImageBind is designed to handle six distinct data types, which encompass the core and emerging elements of modern short-form video content:

- **Image/Video:** The primary visual content, capturing scenes, objects, and events.
- **Audio:** Sound data, including background music, sound effects, and ambient noise, which is critical for setting the tone of a short format.
- **Text:** Written content such as captions, on-screen text, or narrative descriptions that provide context and meaning.
- **Depth (3D):** Information about spatial relationships between objects, enabling a more sophisticated understanding of scene composition for immersive effects.
- **Thermal (heatmap):** Data capturing temperature variations, offering novel ways to segment or understand scenes.
- **IMU (Inertial Measurement Units):** Sensor data recording motion and position, crucial for understanding camera movement and dynamics.

For our project, **image/video, audio, and text** are the primary modalities, but the inclusion of depth and IMU points toward future capabilities like understanding complex motion dynamics or creating 3D-consistent effects for reels.

### 6.6.3 Architecture Overview

While the full framework is not publicly released, the architecture described in Meta AI's research paper outlines a three-component system that provides a strong conceptual model for our work:

- **Modality-Specific Encoders:** Separate encoders (e.g., Vision Transformer for images/video, Audio Spectrogram Transformer for audio) first convert raw data from each modality into a high-dimensional embedding vector. This process extracts the essential features from each input type.
- **Cross-Modal Attention Module:** This component is the intelligence behind the integration. It uses attention mechanisms to weigh the importance of different elements within each modality and then fuses them. For short-form generation, this would allow the system to focus on the most relevant visual element that matches a specific audio cue.
- **Joint Embedding Space:** The final component is the shared vector space where all modality embeddings are projected. This common space enables direct comparison

and combination of concepts from different sources, which is the foundation for tasks like retrieving a video clip that matches an audio track or generating a caption that aligns with a scene.

### 6.6.4 Training Methodology: The "Binding" Property of Images

A key innovation of ImageBind is its efficient training approach. It leverages images as a "binding" modality. The model is trained using **image-paired data (image, X)**, where X can be any one of the other five modalities (text, audio, depth, IMU, or thermal). This means the model learns the relationships between, for example, an image and its corresponding sound, without needing every possible paired combination (e.g., text paired directly with audio). This approach is highly relevant, as it suggests a robust system can be built using widely available image-centric datasets, a principle that can be adapted to training a model for short-form content generation.

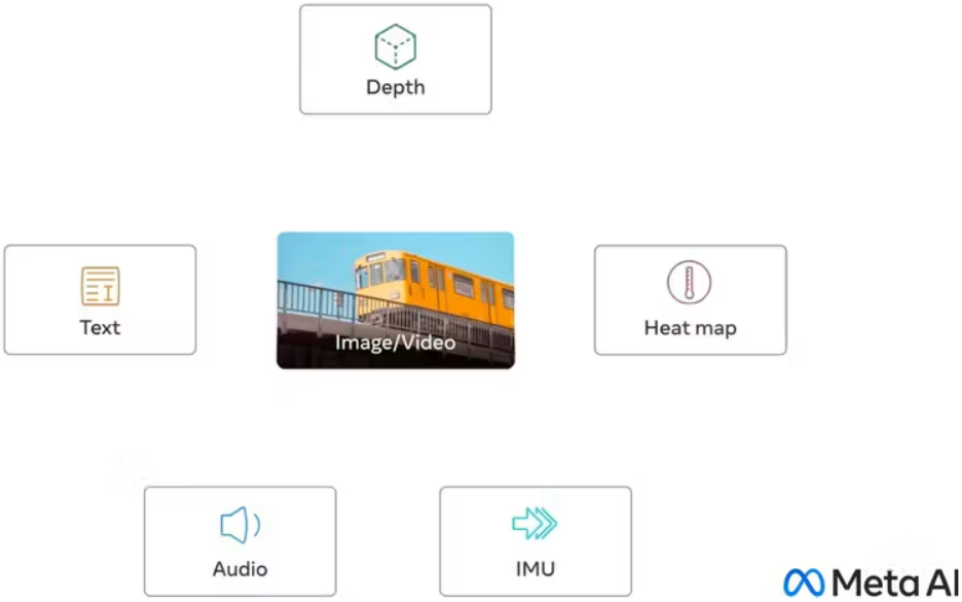


Figure 1.5: ImageBind: Holistic AI learning across six modalities

### 6.6.5 Key Capabilities and Performance Advantages

ImageBind is not merely a theoretical model; it demonstrates superior performance and unique emergent capabilities that are directly relevant to generating high-impact short formats.

- **Outperforming Specialist Models:** ImageBind outperforms prior specialist models

that were trained for a single modality. For example, in few-shot audio classification tasks (using only 1-4 examples), ImageBind achieved approximately **40% higher top-1 accuracy** than self-supervised and even supervised AudioMAE models. Similarly, it outperformed MultiMAE on depth classification tasks. This suggests that a unified model can learn richer, more transferable features than siloed, task-specific models.

- **Emergent Zero-Shot Capabilities:** ImageBind exhibits new scaling behaviorsabilities that don't exist in smaller models but appear in larger versions. This includes:
  - *Cross-Modal Retrieval:* The ability to retrieve content across modalities that were never observed together. For instance, a user could search for a video clip using an audio description like "the sound of a bustling market" or find a text caption that matches a specific thermal image.
  - *Compositional Semantics:* The joint embedding space allows for the addition of embeddings from different modalities to compose their semantics. For our project, this means we could combine the embedding of "a high-energy sports clip" (video) with the embedding of "triumphant orchestral music" (audio) to retrieve or generate a new, cohesive short format that embodies both concepts.
- **Audio-to-Image Generation:** By using its audio embeddings with a pre-trained DALLE-2 decoder (which typically works with CLIP text embeddings), ImageBind enables audio-to-image generation. This capability is a direct precursor to generating entire video sequences from audio prompts, a powerful tool for content creation.

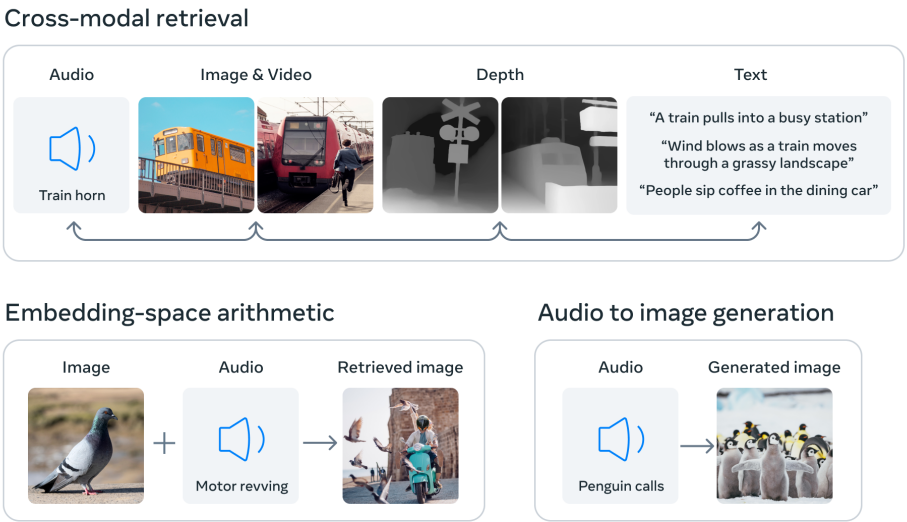


Figure 1.6: ImageBind's joint embedding space enables novel multimodal capabilities

### 6.6.6 Relevance to Automatic Short-Form Content Generation

ImageBind's capabilities directly address the core challenges and open new avenues for our project:

- **Holistic Content Understanding:** For an automated system to generate high-impact reels, it must understand the relationship between a visual scene, its accompanying audio, and the text that narrates it. ImageBind provides the blueprint for a system that can analyze all these elements holistically, not as separate entities.
- **Enhanced Cross-Modal Search and Retrieval:** The joint embedding space allows for powerful cross-modal search. A creator (or an automated system) could build a reel by searching for a video clip using a text prompt like "a serene sunset," then find the perfect matching audio track by using the *video clip itself* as the query. This streamlines the assembly process dramatically.
- **Generative Composition:** The ability to compose embeddings is perhaps the most exciting prospect. This project could leverage this to automatically generate novel short-form sequences. For example, by combining the embedding of a trending visual style with the embedding of a popular audio clip, the system could generate a new reel that is contextually and culturally relevant, maximizing its potential impact.
- **Scalability and Adaptability:** ImageBind's strong scaling behavior shows that as vision models improve (e.g., with models like Meta's DINOv2), the alignment and performance on non-vision tasks like audio classification also improve. This suggests a clear path for enhancing our project's capabilities over time by integrating more powerful foundational models.

### 6.6.7 Limitations and Future Potential

ImageBind's code and model weights are released under a **CC-BY-NC 4.0 license**, which restricts commercial use. Nevertheless, it serves as an exceptional proof-of-concept and blueprint for the field.

The model is part of Meta's broader open-source AI ecosystem, which includes:

- **DINOv2**: A method for training high-performance computer vision models without fine-tuning.

- **Segment Anything (SAM)**: A universal segmentation model for isolating objects in images.

The success of ImageBind demonstrates the potential of multimodal learning for creative applications. It opens possibilities for developing systems that combine multiple modalities, such as creating immersive virtual worlds or enabling rich search across photos, videos, audio, and text.

For our project, ImageBind provides a compelling vision of a future where AI can generate cohesive, high-impact short formats with holistic understanding.

# 7 Comparative Analysis and Discussion

The evolution of video understanding systems reflects a progressive transition from unimodal processing to fully integrated multimodal paradigms. Early approaches relied on unimodal models such as CNNs and LSTMs, where each modalityvision, audio, or textwas analyzed independently. Despite their computing efficiency, these systems were inherently limited in capturing the full semantic richness of video content, effectively operating as "siloed" experts, since significant events often emerge from the interaction of modalities rather than isolated signals[Baltruaitis et al., 2019].

To overcome these limitations, fusion-based approaches were introduced, combining modalities either at the feature level (early fusion) or at the decision level (late fusion) by concatenating features or scores from independent encoders .While this enabled the processing of multi-sensory inputs, it introduced a critical limitation on the quality of alignment : shallow integration often failed to capture deep semantic correlations, leaving the encoders effectively independent and unable to model the rich cross-modal relationships that give video content its meaning [Liu et al., 2024].

The current joint embedding paradigm, exemplified by CLIP and ImageBind, resolved this by mapping heterogeneous modalities into a single vector space, gaining unprecedented zero-shot flexibility and enabling cross-modal retrieval without task-specific training.

However, this progress introduces a new tension: while models like ImageBind extend coverage to up to six modalities, they rely on a central visual backbone to bind other sensors, which can produce alignment errors in unseen combinations. Furthermore,

embedding-based approaches such as CLIP are still limited to static representations, lacking explicit temporal modeling which makes it difficult to capture long-range dependencies in video content . Modern LLM-based approaches like Video-LLaMA further expand modality coverage to 3+ inputs but introduce a trade-off in computational weight, as their generative expressiveness requires significantly more parameters and slower inference than their embedding-only predecessors.

| **Model** | **Paradigm** | **Mod.** | **Zero-shot** | **Temp.** | **Scale** | **Limitation** |
|---|---|---|---|---|---|---|
| CNN/LSTM | Unimodal | 1 | ✗ | Limited | Low | No cross-modal interaction |
| Early/Late Fusion | Fusion | 2 (V+A) | ✗ | Varies | Medium | Independent encoders; weak alignment |
| CLIP | Joint | 2 | ✓ | ✗ | Large | Static image-text only |
| CLAP | Joint | 2 | ✓ | ✗ | Medium | No vision alignment |
| Flamingo | Joint (LLM) | 2 | Few-shot | Partial | Very Large | High computational cost |
| LLaVA | Joint (LLM) | 2 | ✓ | ✗ | Large | Static images only |
| Video-LLaMA | Joint (Video-LLM) | 3+ | ✓ | ✓ | Very Large | Heavy; slow inference |
| ImageBind | Joint | 6 | ✓ | ✗ | Large | No temporal attention |

Table 1.1: Comparative analysis of representative models and paradigms for multimodal video understanding.

From this comparison, it is clear that unimodal and fusion-based systems are insufficient for capturing the complexity of long-form video content, especially when it comes to highlight detection tasks. While joint multimodal models offer a unified semantic representation with strong zero-shot capabilities, they nevertheless have difficulty modeling temporal dynamics, which are crucial for comprehending video narrative structure and event progression.

Among the reviewed methods, embedding-based models such as ImageBind offer strong cross-modal alignment and scalability, making them particularly suitable for retrieval and similarity-based tasks. In contrast, generative models such as Video-LLaMA excel at reasoning over temporal sequences but require significantly more computational resources.

In general, the choice of paradigm depends on the target application: unimodal or fusion techniques may be preferred by lightweight systems, while high-level semantic understanding and cross-modal reasoning benefit from joint multimodal architectures. Despite its present limitations, joint multimodal representation learning offers the best foundation for video highlight identification, where semantic, emotional, and temporal information must be merged.

# 8 Positioning of the Proposed Approach

The trajectory surveyed in this chapter points to a single conclusion: meaningful cross-modal reasoning requires modalities to be aligned at training time, not combined. The approach adopted in this work follows directly from that constraint.

## 8.1 Paradigm Justification

Unimodal approaches, while effective within their domain, fail to capture the semantic richness of video, which is inherently multimodal. Fusion-based methods combine representations learned independently, which limits their ability to model deep cross-modal correspondences and to generalize to scenarios unseen during training.

Joint multimodal representation learning is the paradigm that supports direct tri-modal comparison without modality-specific supervision. By projecting vision, audio, and text into a shared embedding space through contrastive training, it becomes possible to measure cosine similarity across heterogeneous modalities in a consistent manner.

This mechanism is precisely what underpins the scoring strategy adopted in our pipeline: **a segment is considered salient when it exhibits strong joint activity or a significant semantic shift across consecutive modalities**.

## 8.2 Model Justification

Among the joint embedding models surveyed, ImageBind **??** stands out as the most suitable choice for the problem at hand. CLIP 6.1 is limited to the image-text pair and has no audio processing capability. CLAP 6.2 covers audio and text but disregards the visual modality. Video-LLaMA 6.5, while integrating all three modalities, relies on a large language model backbone that makes it ill-suited for dense segment-level inference.

ImageBind provides a unified embedding space trained around vision as the anchor modality, enabling alignment across six modalities without pairwise supervision. For our use case  trimodal embedding extraction over two-second segments at the scale of a full video  this model offers the optimal trade-off between modality coverage, computational compactness, and a well-documented API.

### 8.3 Identified Limitations and Methodological Handling

Two limitations of ImageBind inform the pipeline design. Its lack of temporal attention time being encoded solely through 3D convolutional patches reduces sensitivity to intra-segment dynamics. Silent segments further risk producing uninformative audio embeddings that distort the final score. The adressing of these limitations is detailed in Chapter 2.

## Conclusion

This chapter has traced the evolution of multimodal learning from unimodal baselines to joint embedding spaces, establishing the theoretical foundation for the approach adopted in this work. The analysis converges on ImageBind as the model best suited to the constraints of trimodal highlight detection not as an arbitrary choice, but as the natural endpoint of the progression surveyed. The following chapter translates this grounding into a concrete pipeline design.

# 2

# Design and Implementation

## Introduction

## 1 Implementation Overview

## 2 Preprocessing Pipeline

### 2.1 Video Segmentation

Video segmentation transforms continuous video into smaller temporal units, enabling granular analysis and improved capture of transient events.
The pipeline uses a sliding window mechanism to create overlapping segments rather than processing the entire video. Specifically, a window of two seconds with a one-second stride produces segments such as . In multimodal analysis, where information may cross segment boundaries, this overlap enables the system to record seamless transitions and maintain contextual continuity.

The `compute_segments` function uses `ffprobe` for metadata extraction and applying fixed-precision rounding, to ensure consistent and synchronized segment boundaries, preventing floating-point drift, which could cause misalignment between modalities over long videos.

## 2.2 Frame Extraction

Frame extraction converts each video segment into representative visual samples, then frames are extracted using the `decord` library,for efficient video reading. For each segment, a predetermined number of frames (typically eight) are uniformly sampled between the start and end timestamps. This uniform sampling ensures comprehensive visual representation across the entire segment. Frame indices are calculated by rounding timestamp-to-frame conversions, to maintain temporal accuracy and consistency. The selected frames are then retrieved in batch form and transformed into NumPy arrays, making them compatible with downstream processing.
This approach balances computational efficiency and representational quality by restricting frames per segment while retaining sufficient visual information for multimodal analysis.

## 2.3 Audio Extraction

Audio extraction aims to isolate and structure the acoustic component of each video segment. `ffmpeg` extracts the full audio track, standardized to 16 kHz and mono.
The `librosa` library loads the audio, and segment boundaries signal are converted to sample indices by multiplying with the sampling rate and rounding.This ensures consistent 2-second segments (32,000 samples) and eliminates floating-point inconsistencies.
Each audio segment is thus obtained by slicing the full waveform between two integer indices corresponding to the segment's temporal boundaries. This ensures that all extracted segments have consistent durations (e.g., 2 seconds corresponding to 32,000 samples), which is crucial for stable feature extraction and model input.
Low-level audio features(zero-crossing rate, spectral centroid, RMS energy) are also computed, enhancing multimodal integration.

## 2.4 Text Extraction via Speech Recognition

Text extraction uses automatic speech recognition (ASR), specifically the robust Whisper model, to transcibe spoken content. Transcriptions include precise temoral annotations, enabling alignment with video segments.An alignment procedure concatenates text fragments from the transcription segments that temporally overlap with each video segment, creating a coherent textual representation.
This synchronized trimodal (vision, audio, and text) data is fundamental for deeper semantic analysis and multimodal learning frameworks such as ImageBind ensuring consistency across all modalities.

## 2.5 Modality Synchronization and Alignment

Temporal consistency across modalities is enforced by computing a single canonical set of segment boundaries once via `compute_segments`, using `ffprobe`-derived duration with fixed-precision rounding, and sharing it across all three extraction pipelines. This eliminates floating-point drift that would arise from independent per-modality boundary computation.
Given boundaries $[t_{\text{start}}, t_{\text{end}}]$, each modality maps them to its native domain: frame indices via $\lfloor t \cdot \text{fps} \rceil$ for vision, sample indices via $\lfloor t \cdot f_s \rceil$ for audio, and timestamp intersection for text. This guarantees that $\mathbf{v}_i$, $\mathbf{a}_i$, and $\mathbf{t}_i$ are temporally co-referential for every segment $i$.

# 3 ImageBind Integration

ImageBind [Girdhar et al., 2023] serves as the core multimodal embedding model, extracting unified 1024-dimensional representations from vision, audio, and text modalities into a shared embedding space where cosine similarity measures semantic relatedness across modalities.

## 3.1 Model Configuration

The ImageBind-Huge model is loaded from pretrained weights and deployed in inference mode with frozen parameters. The model employs three modality-specific encoders: a Vision Transforme (ViT-H/14) for frames at 224x224 resolution, a spectrogram encoder for 2-second audio clips converted to 128-bin mel-spectrograms, and a transformer-based text encoder using the CLI tokenizer. All encoders project inputs into the shared 1024-dimensional space via $\ell_2$ normalization and learned temperature scaling.

Listing 2.1: ImageBind initialization

```
import imagebind.models.imagebind_model as ib_model
device = "cuda" if torch.cuda.is_available() else "cpu"
model = ib_model.imagebind_huge(pretrained=True)
model.eval().to(device)
```

## 3.2 Feature Extraction Pipeline

**Vision**: Frames are extracted at the video's native frame rate (24 fps, yielding 48 frames per 2-second segment) using `decord`'s `VideoReader` with a fixed stride of 1, ensuring full temporal coverage without skipping frames.
Each frame is resized to $224 \times 224$ via center-crop, normalized using textttImageNet statistics ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$), and stacked into a tensor of shape $(T \times C \times H \times W)$ before being passed through the vision encoder to produce frame-level embeddings $\mathbf{v}_i \in \mathbb{R}^{1024}$.

**Audio**: Audio tracks are demuxed from the video container via FFmpeg and resampled to 16 kHz mono to match ImageBind's expected input format. Each 2-second clip is time-aligned with its corresponding video segment using a shared duration reference computed once via `ffprobe`, eliminating cross-modality boundary drift. Clips are converted to 128-bin log-mel spectrograms with a hop length of 10 ms and encoded to produce segment-level embeddings $\mathbf{a}_i \in \mathbb{R}^{1024}$.

**Text**: Whisper-large-v3 transcribes the full audio stream with word-level timestamps, after which transcriptions are mapped to their corresponding segments by timestamp intersection. Transcriptions are tokenized via the CLIP tokenizer with a maximum context of 77 tokens; segments exceeding this limit are truncated at the token boundary. The token

sequence is encoded to produce $\mathbf{t}_i \in \mathbb{R}^{1024}$.
Silent or non-speech segments initially receive zero-valued embeddings $\mathbf{t}_i = \mathbf{0}$, which are subsequently replaced by the text imagination module( Section 4).

## 3.3 Temporal Aggregation

ImageBind processes frames independently, producing one embedding per frame rather than per segment. To obtain a single segment-level vision representation, mean pooling is applied across the $T$ frame embeddings within each segment:

$$\mathbf{v}_{\text{seg}} = \frac{1}{T} \sum_{j=1}^{T} \mathbf{v}_j \tag{2.1}$$

The result is re-normalized to preserve unit norm and maintain consistency with ImageBind's $\ell_2$-normalized embedding geometry:

$$\mathbf{v}_{\text{seg}}^{\text{norm}} = \frac{\mathbf{v}_{\text{seg}}}{\|\mathbf{v}_{\text{seg}}\|_2} \tag{2.2}$$

Audio and text encoders produce segment-level embeddings directly and require no further aggregation. Mean pooling was retained over attention-based alternatives for its determinism and computational simplicity.

## 3.4 Embedding Storage

Per-video embeddings are persisted to compressed NumPy archives (`.npz`), one file per video, containing the three modality matrices and associated metadata:

Listing 2.2: Embedding storage schema

```
{
  'vision':          np.float32,  # (N, 1024)
  'audio':           np.float32,  # (N, 1024)
  'text':            np.float32,  # (N, 1024)
  'timestamps':      np.float32,  # (N, 2)  [start, end] in sec
  'modality_valid': {             # validity flags per-segment
      'vision': bool[],
      'audio':  bool[],
```

```
      'text':   bool[]
  }
}
```

The `modality_valid` flags record whether each modality produced a meaningful embedding for a given segment. A text flag of `False` indicates a silent or non-speech segment whose embedding is temporarily set to the zero vector $\mathbf{t}_i = \mathbf{0}$, pending replacement by the text imagination module (Section 4), after which ImageBind re-encodes the imputed text to produce a valid $\mathbf{t}_i \in \mathbb{R}^{1024}$.

# 4 Missing Modality Handling: Text Imagination

# 5 Highlight Scoring and Selection

# 6 Evaluation Metrics and Validation Protocol

# Conclusion and perspectives

# A

# Appendix 1

Insert your appendixes here if you need.

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